

AIR QUALITY COMPARISON: MEXICO'S LITERATURE REVIEW

Week 4 – Dust, O₃, Health & Weather

- **Objective:** connect Saharan dust events, ozone policy, health exposure, and meteorological drivers in Mexico City.
- **Focus:** literature synthesis rather than new data.
- **Framework:** Event → Policy → Health → Patterns → Integration.
- **'Week' 4 period:** Sep. 27th – Oct. 17th 2025.



SAHARAN DUST TRANSPORT TO MEXICO (CONTEXT)

Raga et al. (2021)

- African dust plumes reach Mexico every summer (June–July).
- Main transport path: trans-Atlantic → Caribbean → Gulf of Mexico → Yucatán → interior of Mexico.
- Dust raises background PM levels across the country, even without local emissions.
- Provides context for Mexico City summer PM peaks.

Saharan dust regularly reaches Mexico each summer, raising the PM baseline even before local emissions act.

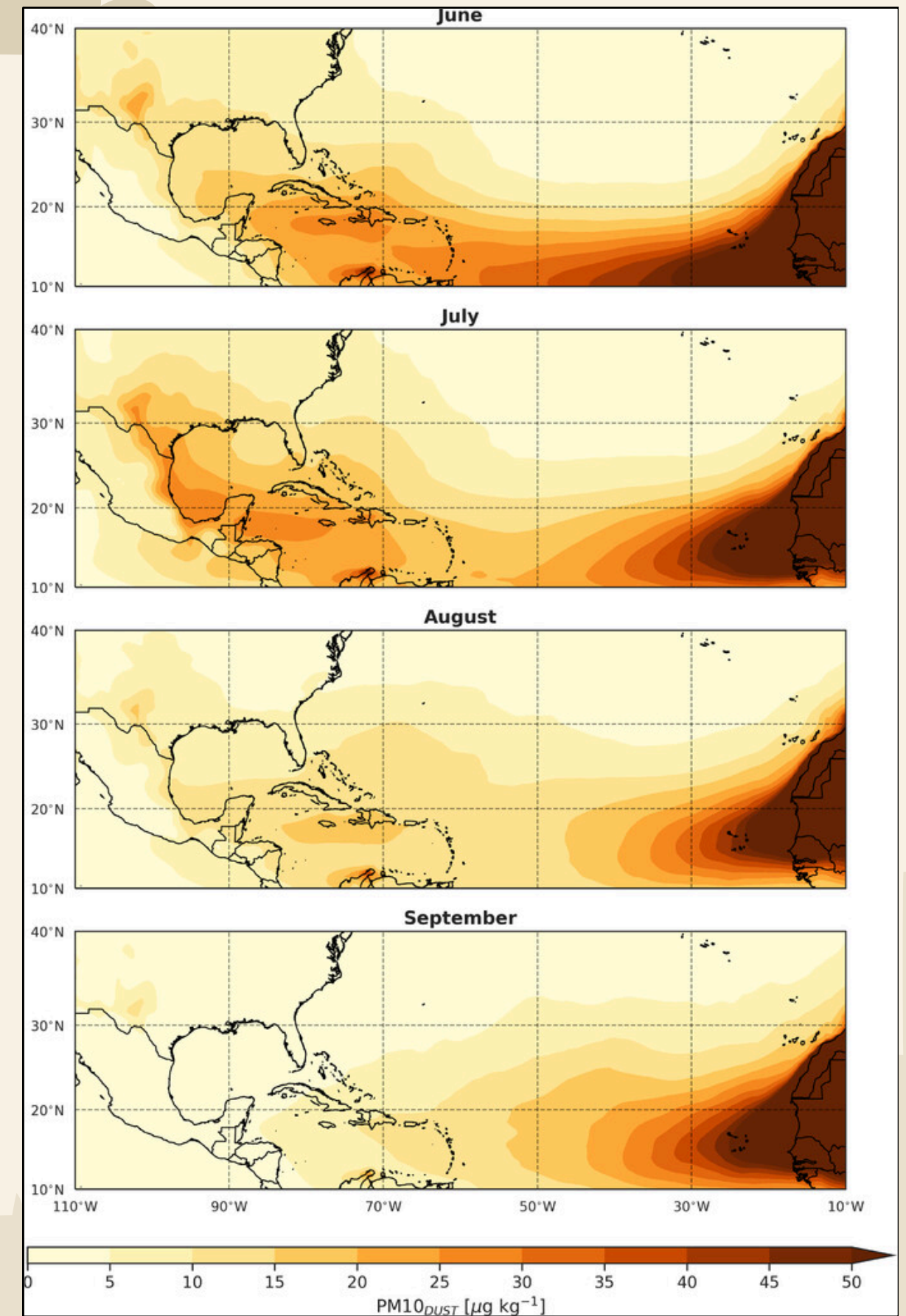
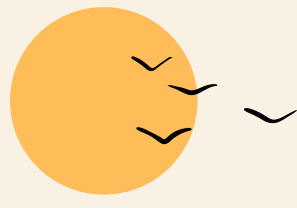


Fig. 1. Mean monthly PM₁₀ concentration associated with dust derived from MERRA-2 over 2003–19, for June–September. (Raga et al., 2021)



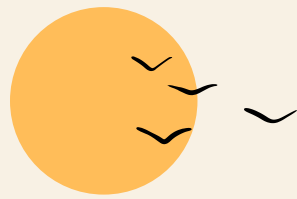
JUNE 2020 DUST EVENT: PM RISE & HEALTH RISK (1/2)

Kutralam-Muniasamy et al. (2021)

- Research work in other Mexican regions.
- June 2020 event: PM₁₀ ↑ to 197 µg/m³; PM_{2.5} ↑ to 94 µg/m³ (2–5x increase).
- Concentrations exceeded Mexican NOM-020-SSA1-2014 limits (old NOM).
- Positive correlation between PM levels and COVID-19 cases and deaths.
- Healthrisk assessment: higher chronic daily intake and hazard quotient during dust days.
- Limitation: meteorological factors not included.



Fig. 2. Own adaptation using MapChart.



JUNE 2020 DUST EVENT: PM RISE & HEALTH RISK (2/2)

Kutralam-Muniasamy et al. (2021)

The 2020 dust intrusion doubled or quintupled PM levels, clearly demonstrating the health relevance of trans-Atlantic events.

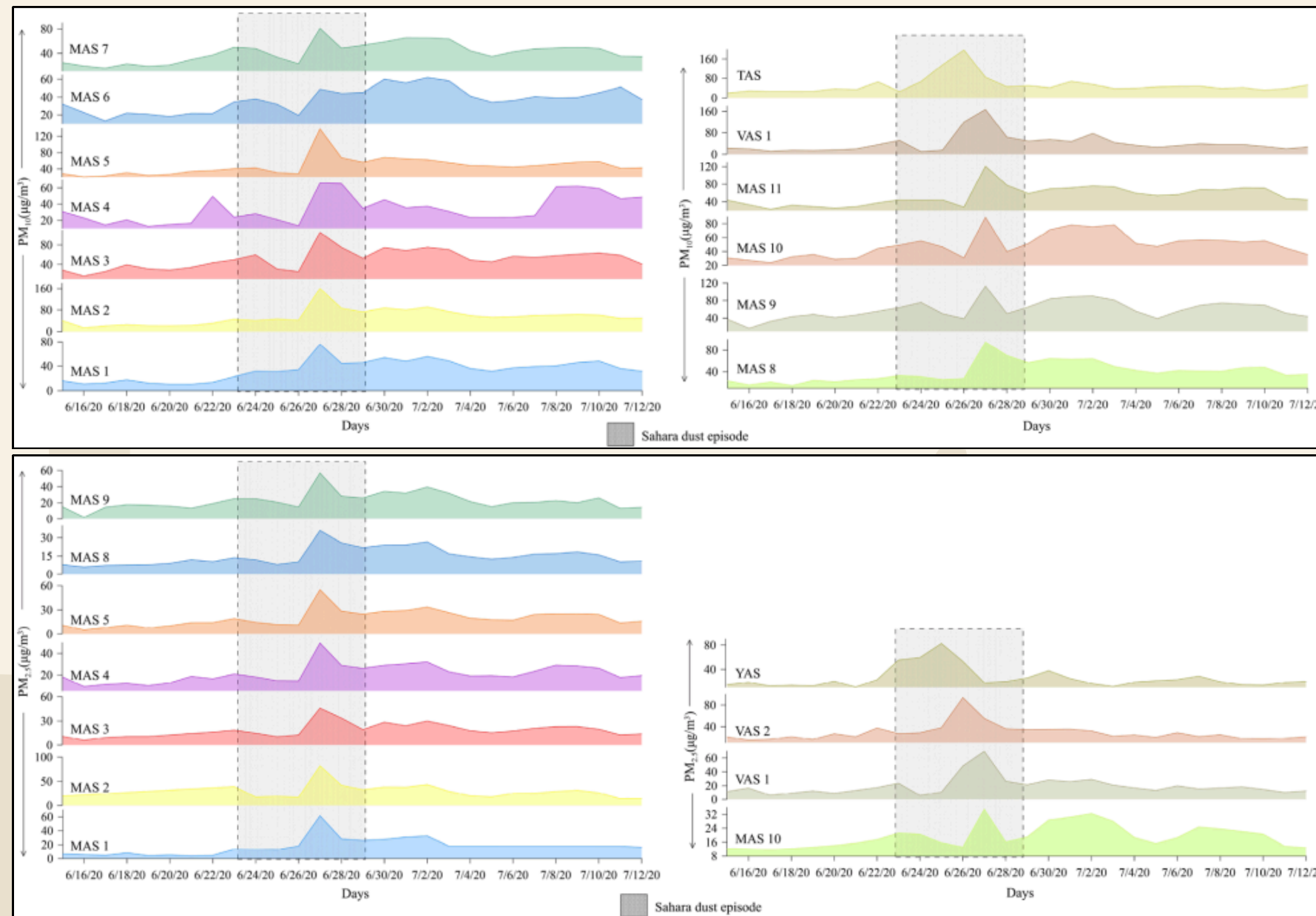


Fig. 4. Average PM₁₀ and PM_{2.5} concentrations from June 15 to July 12, 2020 recorded in air monitoring stations located in Nuevo Leon, Veracruz, and Tabasco of Mexico. (Kutralam-Muniasamy et al. 2021).

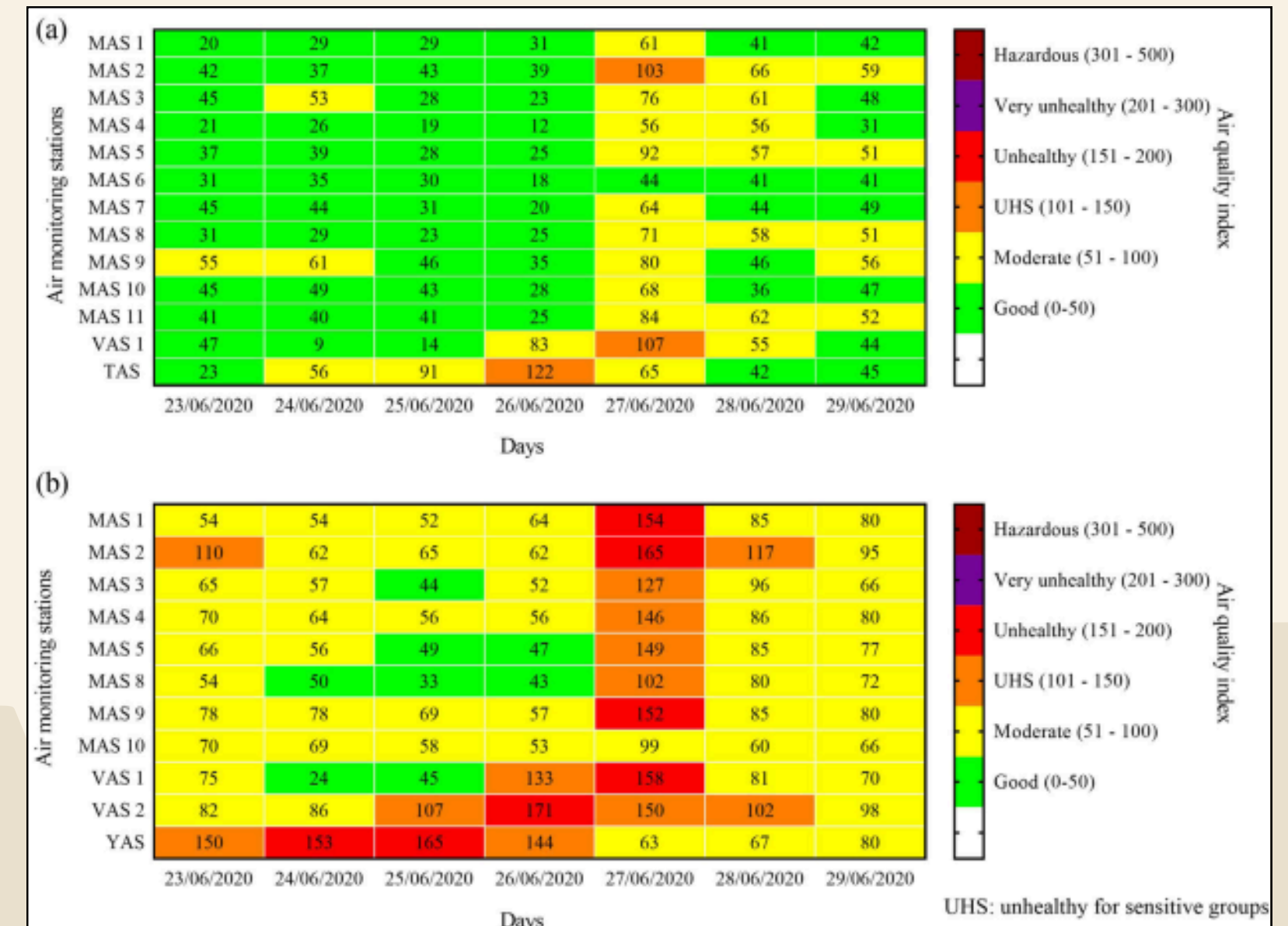


Fig. 3. AQI levels for PM₁₀ and PM_{2.5} concentration for the period of assessment (June 23–29, 2020) (Kutralam-Muniasamy et al. 2021).

FROM DUST EVENTS TO POLICY: WHY 1-HOUR O₃ MATTERS

SEMARNAT (2023)

- Frequent acute events → need for clear public communication.
- NOM-172 (SEMARNAT, 2023) removes 8-hour O₃ metric; keeps 1-hour average to avoid contradictory messages.
- **Scientific basis:** NOM-172 was based on Peng et al. (2013) acute O₃ mortality study + 1.14 conversion ($O_{3-1h} = 1.14 \times O_{3-8h}$) from Anderson & Bell (2010).
- Aligns with 2021 WHO guidelines on short-term exposure and risk communication.

Mexico adopted the 1-hour ozone index to communicate acute health risk more clearly and consistently with WHO evidence.

“

The implementation of the AIR and HEALTH Index has posed a challenge for effective communication in the specific case of ozone, as it contemplates the simultaneous use of two indicators of air quality. One to communicate the state of air quality and the associated health risks from exposure to 1-hour average concentrations, and the other to communicate the state of air quality and the associated health risks from exposure to 8-hour average concentrations. This situation frequently leads to the need to communicate to the population, for a single hour, that the air quality condition may be poor, very poor, or extremely poor for one indicator, while for the other, the condition may reflect good or acceptable air quality. This creates confusion among the public about making the best decisions to reduce their exposure to high concentrations of this pollutant. For this reason, only the one-hour index will be applied.

“

For ozone, the study by Peng et al. (2013) was used with a 1-hour exposure metric, and the O₃-1hr = O₃-8hr ratio of 1.14 was used for the calculations (Anderson and Bell, 2010).

SHORT-TERM EXPOSURE & CARDIOVASCULAR IMPACTS (1/2)

Ugalde-Resano et al. (2021)

Air-pollution spikes translate into immediate cardiovascular stress, about 10 % of emergency visits are pollution-related.

- 48,891 cardiovascular emergency visits (2016–2019) in Mexico City.
- Air pollutants were responsible for 9.7% to 11% of all emergency department visits (CEDVs) for cardiovascular and respiratory diseases.
- **China comparison:** Similar studies show particulate matter (PM) and gases are linked to 3.3% to 6% of general emergency department visits.

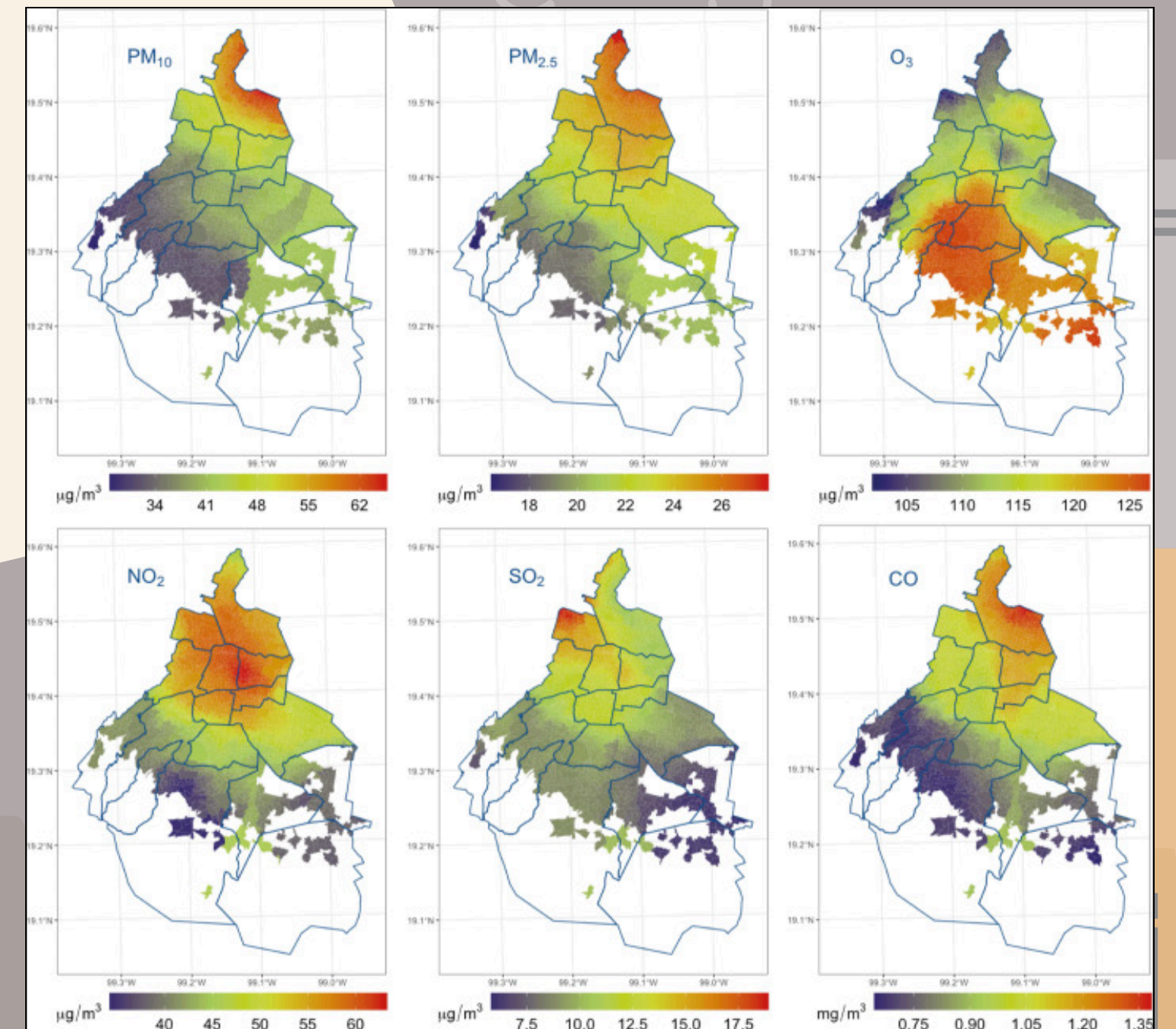
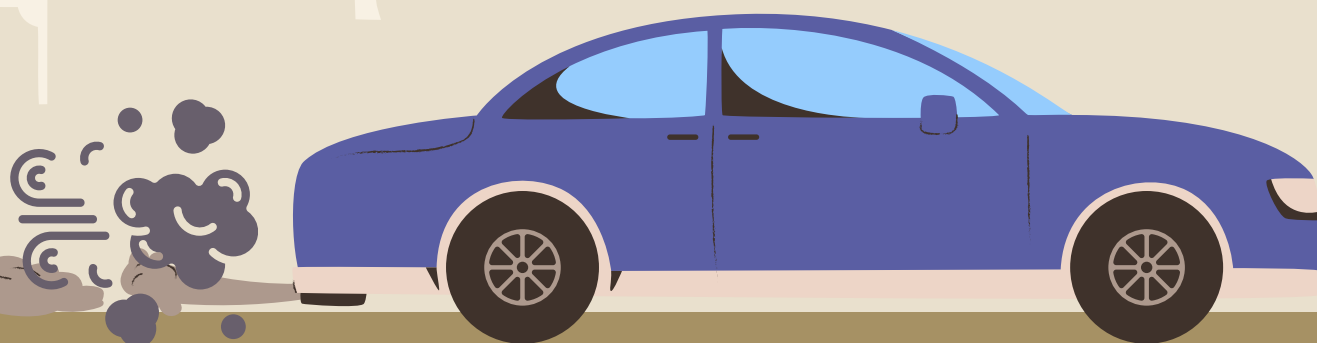


Fig. 4. Spatial distribution of daily average concentrations of air pollutants in Mexico City (16/05/2016–28/02/2019) (Ugalde-Resano et al., 2022).



SHORT-TERM EXPOSURE & CARDIOVASCULAR IMPACTS (2/2)

Ugalde-Resano et al. (2021)

- Per 10 $\mu\text{g}/\text{m}^3$ increase: PM10 +2.8 %, PM2.5 +3.7 %, O₃ +1.1 %, NO₂ +2.5 %; CO (1 mg/m³) +6.6 %.
- Average lag 0–5 days → effects appear within a week after exposure.
- Data demonstrates a significant and quantifiable link between common air pollutants and the demand for emergency health services in large urban centers → acute health alerts for pollution spikes.

In environmental health studies, lag = delay in days between exposure and the observed effect.

- Lag 0 = same day as the peak of contamination.
- Lag 1 = effect the following day, etc.
- Lag 0–3 = average cumulative exposure over 4 days.

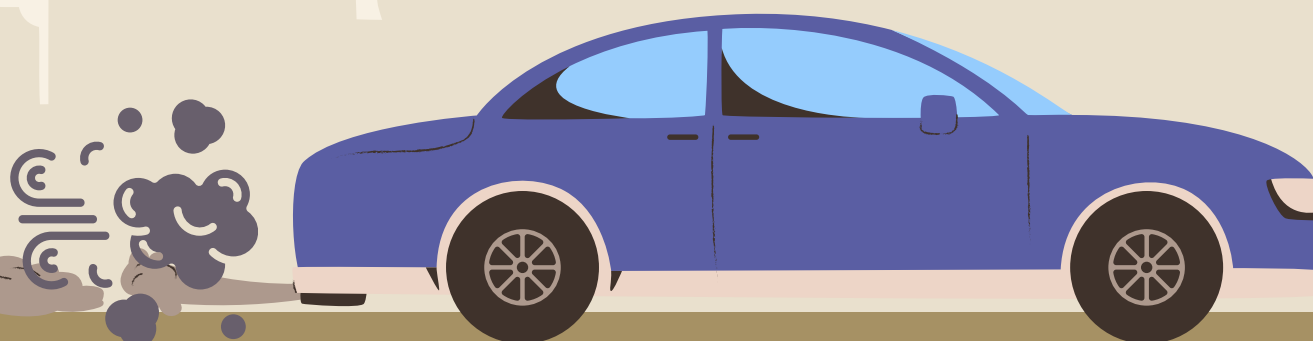


Table 1. Statistical description of daily cardiovascular emergency department visits, ambient air pollutants and weather conditions in Mexico City by season (16/05/2016–28/02/2019) (Ugalde-Resano et al., 2022).

	Dry-warm mean $\pm$ SD	Rainy mean $\pm$ SD	Dry-cold mean $\pm$ SD	p ^a
Cardiovascular emergency department visits				
All	49 $\pm$ 16	49 $\pm$ 16	46 $\pm$ 16	0.07
Male	22 $\pm$ 9	22 $\pm$ 9	21 $\pm$ 8	0.40
Female	27 $\pm$ 9	27 $\pm$ 9	25 $\pm$ 9	0.025*
30 to 59 years	24 $\pm$ 9	24 $\pm$ 9	22 $\pm$ 8	0.017*
≥ 60 years	21 $\pm$ 7	21 $\pm$ 8	21 $\pm$ 8	0.78
HBP	32 $\pm$ 10	32 $\pm$ 10	30 $\pm$ 9	0.006*
IHD	4 $\pm$ 2	4 $\pm$ 3	4 $\pm$ 3	0.07
CVA	6 $\pm$ 3	5 $\pm$ 3	5 $\pm$ 3	0.86
Ambien air pollutants				
PM ₁₀ ($\mu\text{g}/\text{m}^3$)	48 $\pm$ 11	31 $\pm$ 10	52 $\pm$ 16	<0.001*
PM _{2.5} ($\mu\text{g}/\text{m}^3$)	27 $\pm$ 7	17 $\pm$ 6	27 $\pm$ 10	<0.001*
O ₃ ($\mu\text{g}/\text{m}^3$)	145 $\pm$ 30	108 $\pm$ 32	111 $\pm$ 31	<0.001*
NO ₂ ($\mu\text{g}/\text{m}^3$)	51 $\pm$ 9	44 $\pm$ 10	58 $\pm$ 15	<0.001*
SO ₂ ($\mu\text{g}/\text{m}^3$)	9 $\pm$ 8	7 $\pm$ 6	15 $\pm$ 14	<0.001*
CO (mg/ m ³)	1.03 $\pm$ 0.30	0.82 $\pm$ 0.24	1.16 $\pm$ 0.36	<0.001*
Weather conditions				
Temperature (°C)	18.8 $\pm$ 2	17.7 $\pm$ 1	15.3 $\pm$ 2	<0.001*
Relative humidity (%)	46 $\pm$ 12	64 $\pm$ 8	50 $\pm$ 12	<0.001*

SD: standard deviation; Dry-warm: March-May, Rainy: June-October and Dry-cold: November-February; Cardiovascular emergency department visit: CIE-10: I00-I99, HBP (High blood pressure): CIE-10: I10-I16, IHD (Ischemic heart disease): CIE-10: I20-I25, CVA (Cerebrovascular accident): CIE-10: I60-I69; air pollutants were estimated using the 24-hour daily average for PM₁₀, PM_{2.5}, NO₂ and SO₂, and the maximum value of the 8-hour moving average for O₃ and CO. ^a ANOVA, *p<0.05.



LONG-TERM EXPOSURE & COVID-19 MORTALITY

(1/3)

López-Feldman et al. (2021)

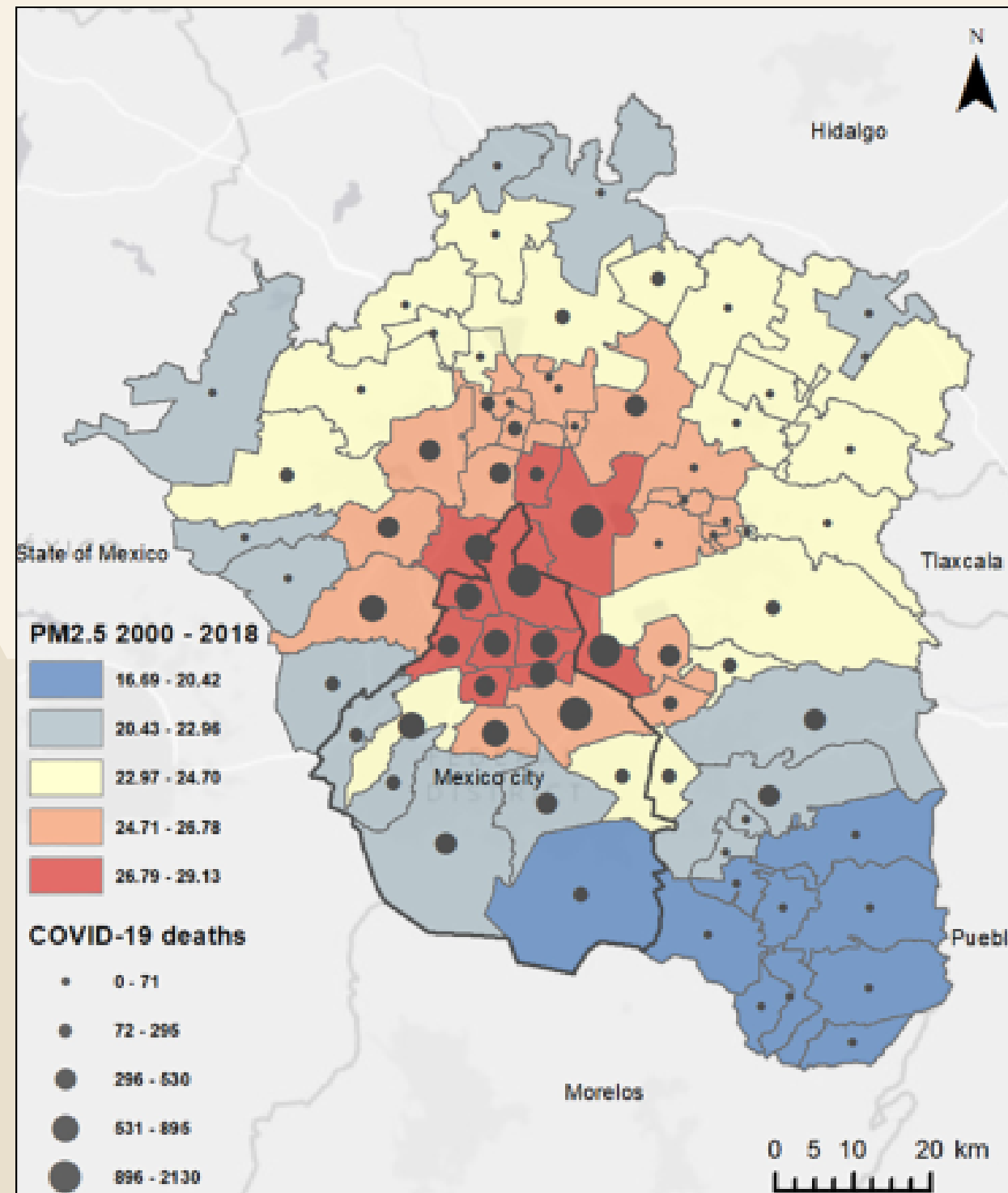


Fig. 5. Distribution of long-term pollution (PM_{2.5}) and COVID-19 deaths in Mexico City and the Mexico City Metropolitan Area (MCMA) (López-Feldman et al., 2021).

- Individual-level Mexico City Metropolitan Area (MCMA) study (2021).
- Long-term PM_{2.5} exposure significantly increases COVID-19 mortality risk.
- Stronger effects in older populations (≥ 60 years).
- Short-term exposure less relevant than chronic load.
- Demonstrates how air pollution amplifies pandemic vulnerability.

All municipalities in the Mexico City Metropolitan Area exceed the WHO annual PM standard, with the highest exposure concentrated in dense, economic hubs, suggesting a correlation with elevated COVID-19 mortality rates in those same areas.



LONG-TERM EXPOSURE & COVID-19 MORTALITY

(2/3)

López-Feldman et al. (2021)

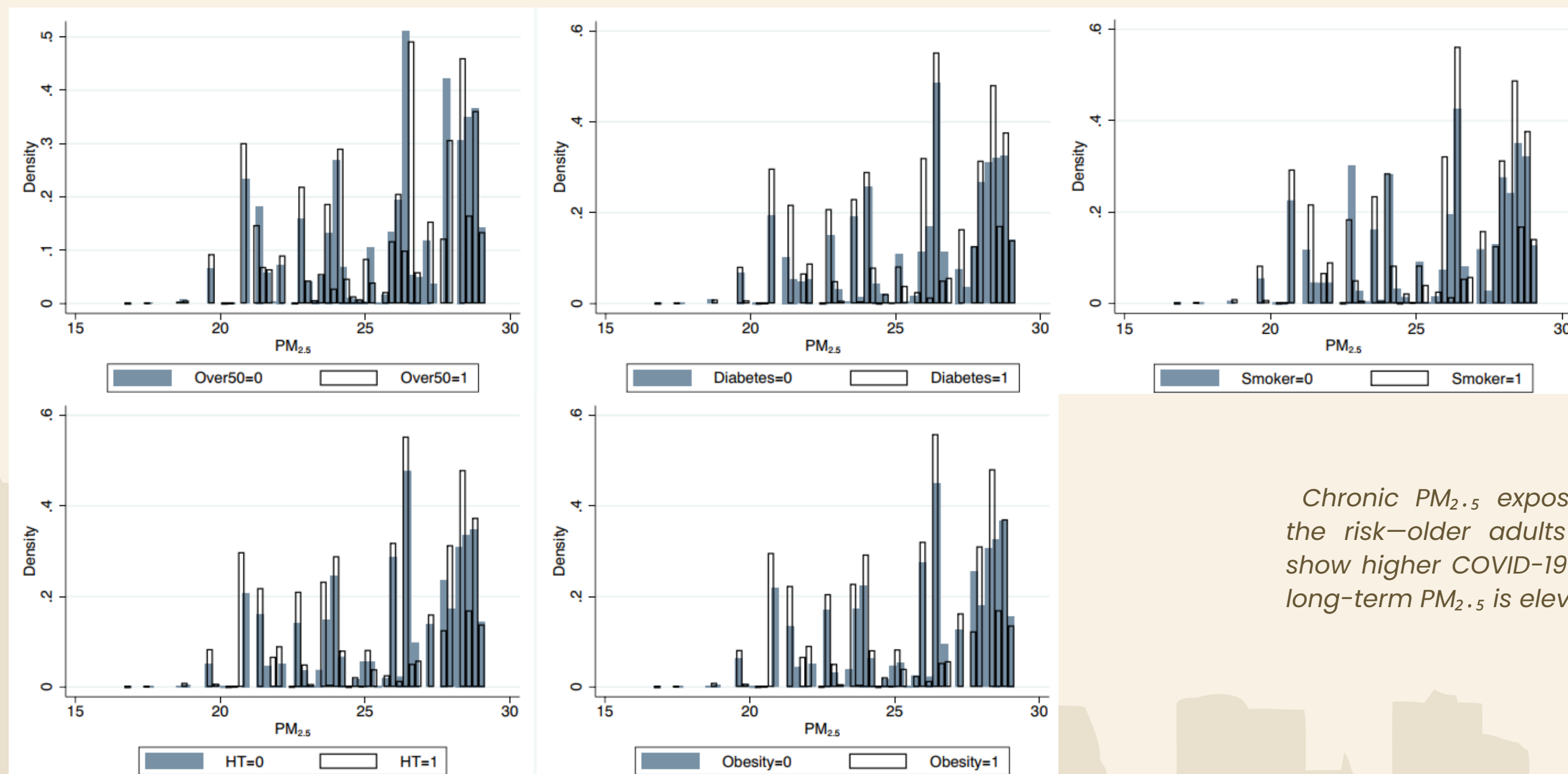


Fig. 6. Distribution of long-term exposure to $PM_{2.5}$ in the MCMA by individual characteristics. Each panel shows a histogram of the average concentration of $PM_{2.5}$ to which individuals grouped by a given category were exposed. The categories are: Over50 (a variable equal to one if an individual is at least 50 years old); Diabetes (=1 if the individual was diagnosed with type-2 diabetes); HT (=1 if the individual was diagnosed with hypertension); Obesity(=1 if the individual was diagnosed as obese); and Smoker (=1 if the individual smokes) (López-Feldman et al., 2021).

Chronic $PM_{2.5}$ exposure compounds the risk—older adults in Mexico City show higher COVID-19 mortality where long-term $PM_{2.5}$ is elevated.





LONG-TERM EXPOSURE & COVID-19 MORTALITY

(3/3)

López-Feldman et al. (2021)



Table 2. The relationship between long- and short-term exposure to pollution and the probability on dying from COVID-19 in Mexico City (López-Feldman et al., 2021).

	Model 1	Model 2	Model 3
PM _{2.5} (µg/m ³)	0.0424*** [0.0143]	0.0733*** [0.0096]	0.0571*** [0.0093]
Municipal-level covariates included	No	Yes	Yes
Individual-level covariates included	No	No	Yes
N	196,273	196,273	196,273
Municipalities	76	76	76
Pseudo-R ²	0.006	0.027	0.265

Notes: The dependent variable is a dummy equal to one if an individual diagnosed with COVID-19 dies and zero otherwise. Estimations are done using a probit model [...]. The Mexico City sample only considers the 14 municipalities for which, in addition to long-term exposure to PM2.5, we know PM2.5 for 2019 and 2020.

- * $p < 0.1$.
- ** $p < 0.05$.

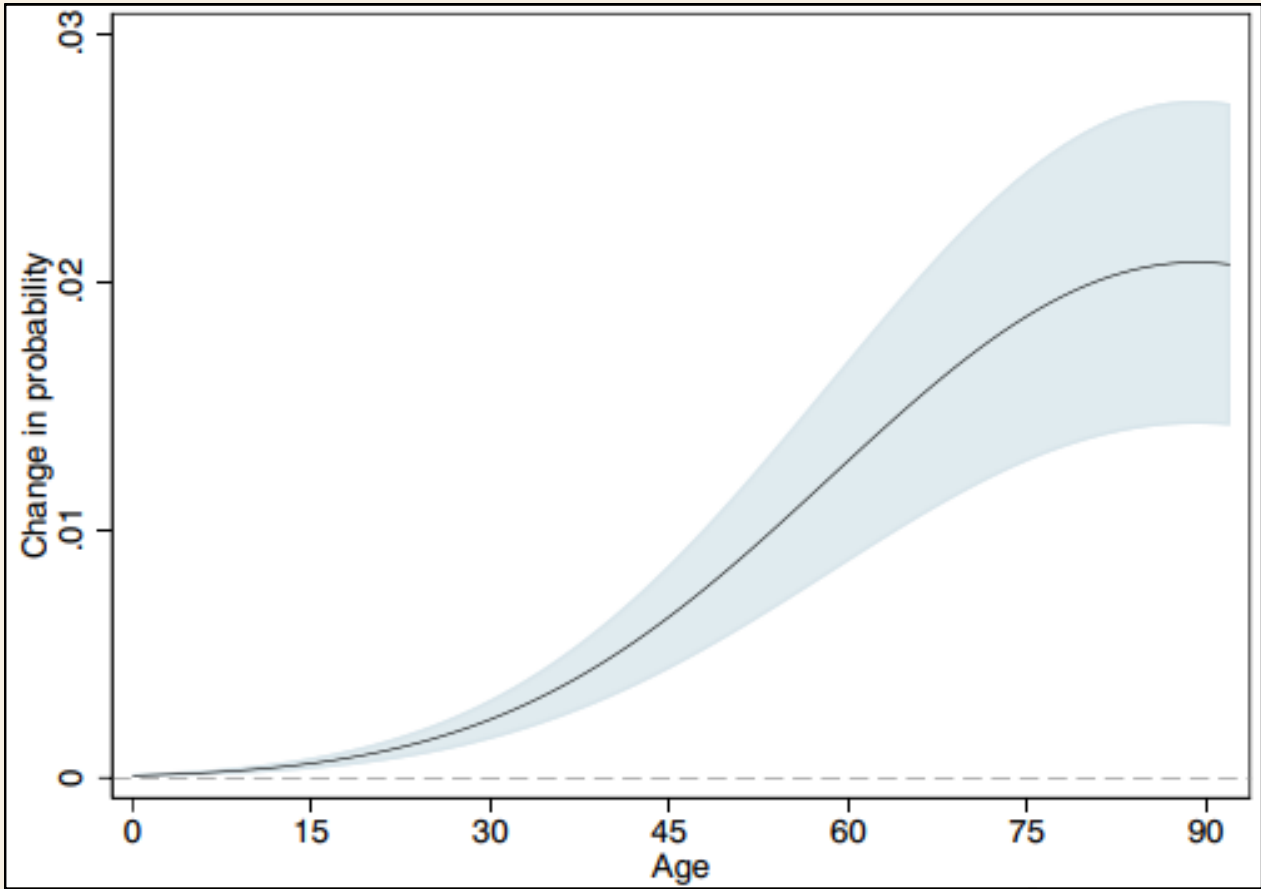
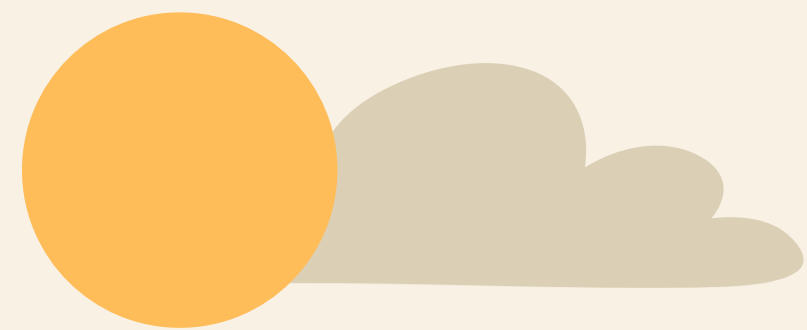


Fig. 7. The effects of exposure to PM2.5 air pollution on probability of COVID-19 death by age in the MCMA. Estimates of marginal effects to exposure to PM2.5 (2000–2018) shown with 95% cluster-robust confidence intervals. [...]. (López-Feldman et al., 2021).

Model 1 uses short-term and long-term pollution exposure; Model 2 replaces the long-term variable with PM2.5–2019; and Model 3 includes all three pollution variables (short-term, original long-term, and PM2.5–2019) to test for the substitutability of the long-term variables.





BUILDING A HEALTH-BASED INDEX

Cromar et al. (2021)

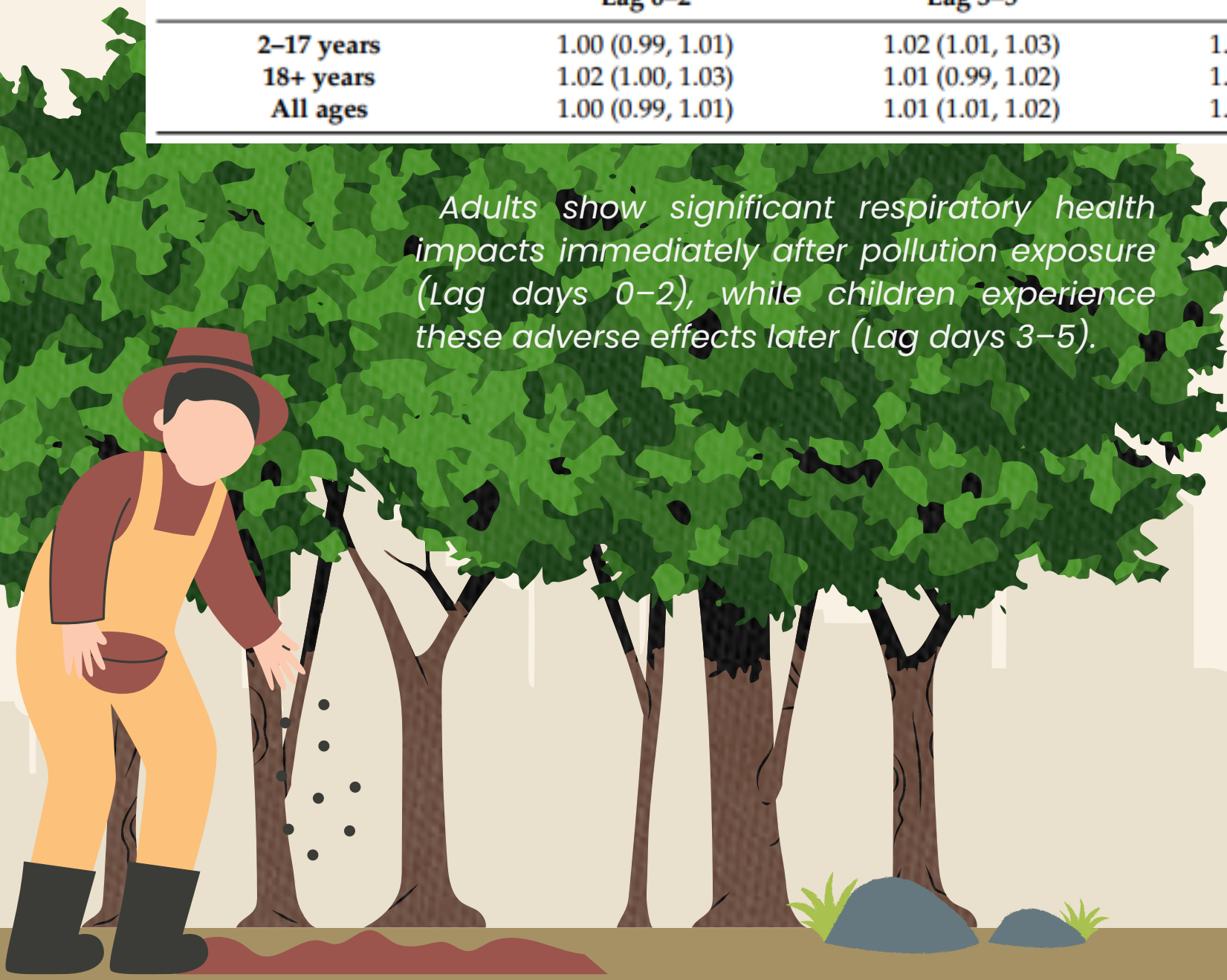
Table 3. Risk ratios for respiratory emergency department visits in Mexico City associated with health-based index values, by age group and lag structure. The primary exposure period of interest is the 6-day average observed at lag days 0–5 [...]. (Cromar et al., 2021).

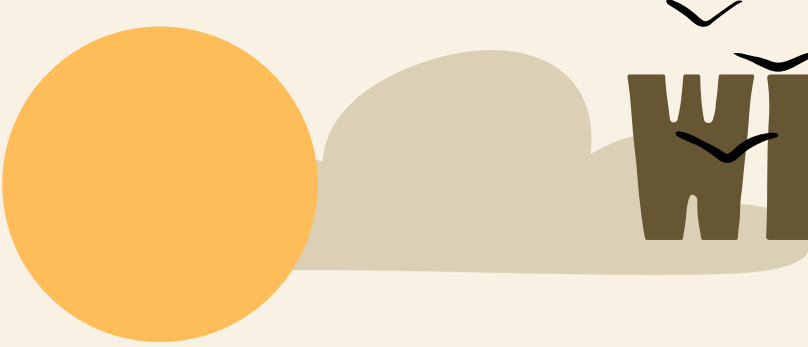
Age	Health-Based Index Values		
	Risk Ratio (95% CI)		
	Lag 0–2	Lag 3–5	Lag 0–5
2–17 years	1.00 (0.99, 1.01)	1.02 (1.01, 1.03)	1.02 (1.00, 1.03)
18+ years	1.02 (1.00, 1.03)	1.01 (0.99, 1.02)	1.02 (1.00, 1.04)
All ages	1.00 (0.99, 1.01)	1.01 (1.01, 1.02)	1.02 (1.00, 1.03)

Adults show significant respiratory health impacts immediately after pollution exposure (Lag days 0–2), while children experience these adverse effects later (Lag days 3–5).

- Multi-pollutant index based only on PM2.5, O3, NO2 + 610,982 ED visits (2010–2015).
- Relative risks: PM2.5 – 1.03 (95 % CI 1.01–1.04); O3 – 1.03; NO2 – 1.01.
- Valid for children and adults; lag 0–3 days captures peak effects.
- Index implemented by SEDEMA as “Conoce tu Número” (“Known your Number”) (2021).
- Foundation for current NOM- 172 index.

Cromar et al. (2021) built a multi-pollutant health-based index (with a guide on how to calculate the values), later adopted by SEDEMA as ‘Conoce tu Número’, integrating local hospital data to daily air quality reporting.





WEATHER PATTERNS: LOCAL FLOW

(2001–2010)

Salcido et al. (2019)

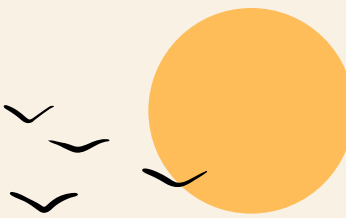
Table 4. Wind Patterns description (Salcido et al., 2019).

This Work	Description
WIND-P1	Early morning downslope winds
WIND-P2	Northeasterly and easterly winds
WIND-P3	Midday northerly winds
WIND-P4	Westerly winds at the western quadrants and Southerly winds at the eastern quadrants
WIND-P5	Afternoon northerly winds
WIND-P6	Midnight downslope winds
WIND-P7	UHI-driven winds

Topography and nighttime convergence trap pollutants; local wind regimes explain where accumulation occurs.

- Salcido (2019): 7 wind circulation in total + 9 pollutant flow patterns for each of the four major pollutants (NO₂, O₃, PM₁₀, SO₂).
 - Modeled using 1-h RAMA data + wind vectors (k-means).
 - Orography and nighttime convergence → pollutant accumulation.
 - Connects surface flow with larger-scale weather patterns.
- 

SYNOPTIC DRIVERS OF O3 CONTINGENCIES



Silva-Quiroz et al. (2019)

- 2004–2018 data → O₃ extreme days (percentile 90) linked to 500 hPa blocking events.
- March–April–May (dry-hot season): low winds, low RH, weak mixing = O₃ build-up.
- O₃ events linked to ↑ T + air stagnation.
- Extreme ozone days → composite anomalies show +110 m Z500 (500 hPa).
- Rossby-wave breaking mixes the atmosphere, forcing naturally high-ozone, high-PV stratospheric air down into the troposphere → combined with local pollution → extreme O₃ events.
- Confirms meteorology as trigger for environmental contingencies.

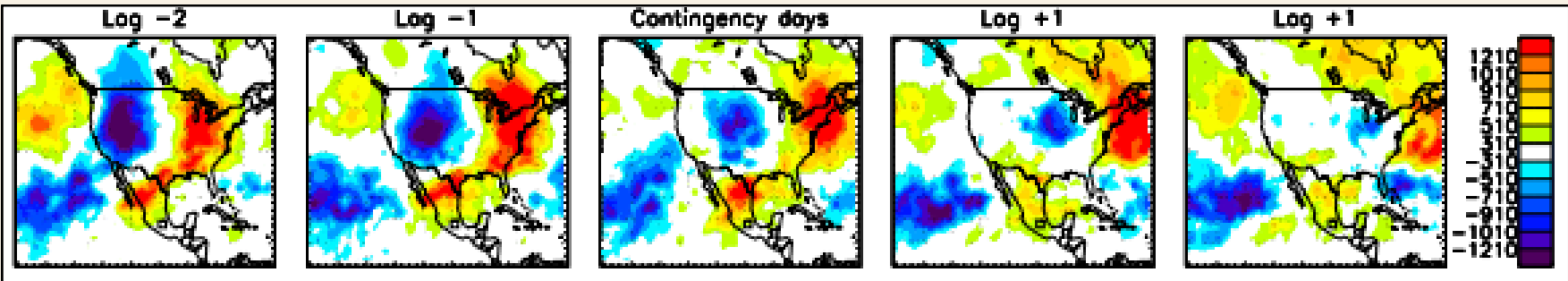


Fig. 8. Composites of anomalies of potential temperature on the tropopause (-2 PVU surface) for extreme ozone events and 2 days before and 2 days after these events. (Silva-Quiroz et al., 2019).

Table 5. Average values of atmospheric composition and meteorological variables during environmental contingency and normal days (Silva-Quiroz et al., 2019).

Variable	Environmental contingencies	Normal days
NO _x (ppb)	92 ± 40	89 ± 49
Ozone (ppb)	81 ± 12*	45 ± 19
Surface Wind Speed (m/s)	1.76 ± 0.09*	3.1 ± 2
500 hPa Wind Speed (m/s)	5.5 ± 2.3	6.4 ± 4.6
Relative Humidity (%)	48 ± 7*	38 ± 12
Temperature (°C)	25 ± 2*	17 ± 9

* Indicates statistically significant difference.

Blocking highs and stagnant air in spring trigger ozone exceedances: meteorology is the key driver of contingencies.



WEATHER PATTERNS & SEASONALITY (1/3)

Díaz-Esteban et al. (2022)

- 30-year (1990–2019) cluster of 500 hPa winds → 7 weather patterns.
- O₃ peaks after noon; PM_{2.5} peaks during morning; NO₂ and PM₁₀ peaks in rush hours.
- La Niña years show more pollution-related patterns than El Niño's.
- Provides predictability for sub-seasonal air quality assessment.
- Core reference for linking meteorology and pollution episodes.

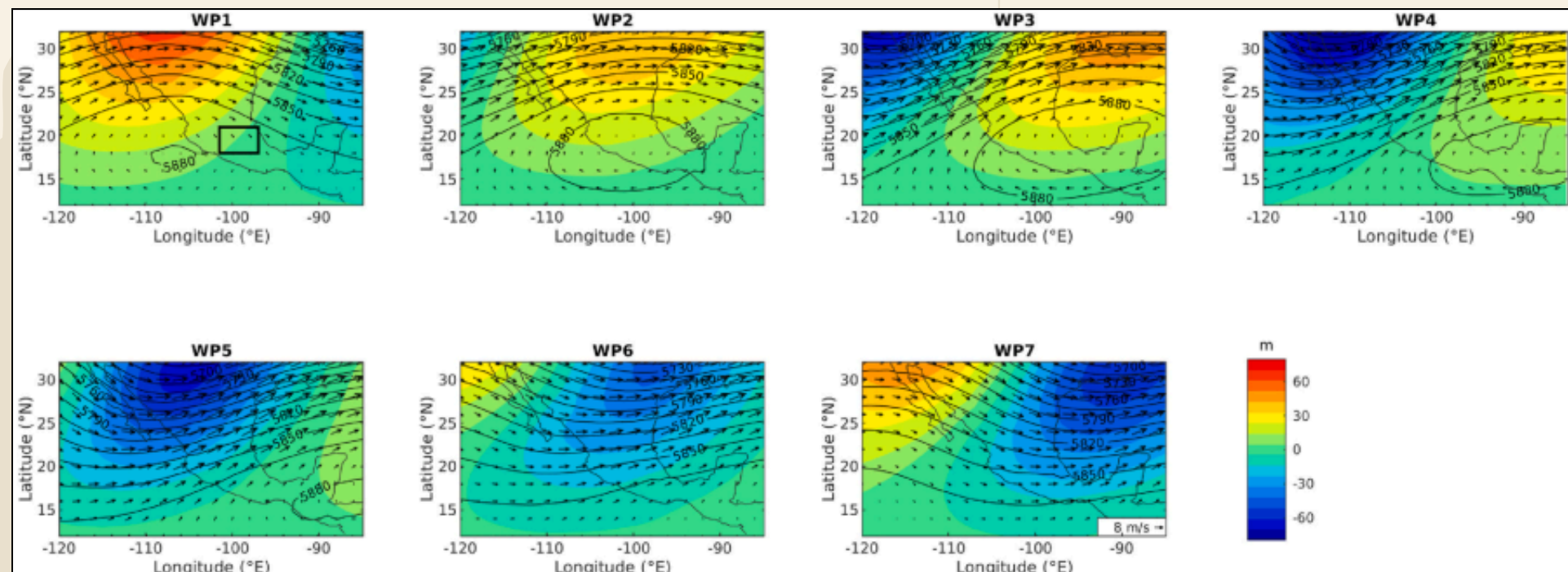
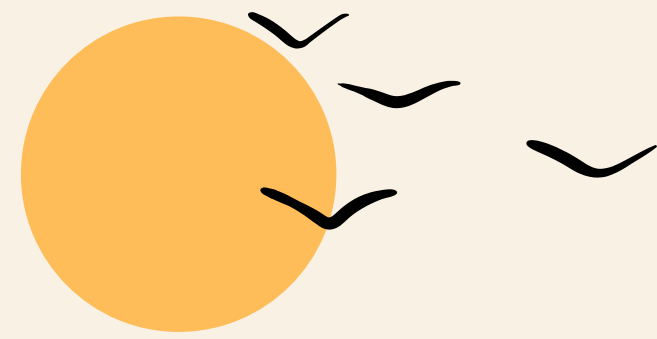


Fig. 9. Composites of mean 500-hPa geopotential height (contours, in m), their anomalies (shading, in m) and mean 500-hPa wind (vectors, in m/s), for each weather pattern. Box in WP1 defines the Mexico City Metropolitan Area (Díaz-Esteban et al., 2019).



Weather pattern analysis from Díaz-Esteban 2022 shows that high-pressure systems dominate O₃ peaks in spring, while NO₂ and PM_{2.5} events occur in winter; La Niña years bring more of these patterns.

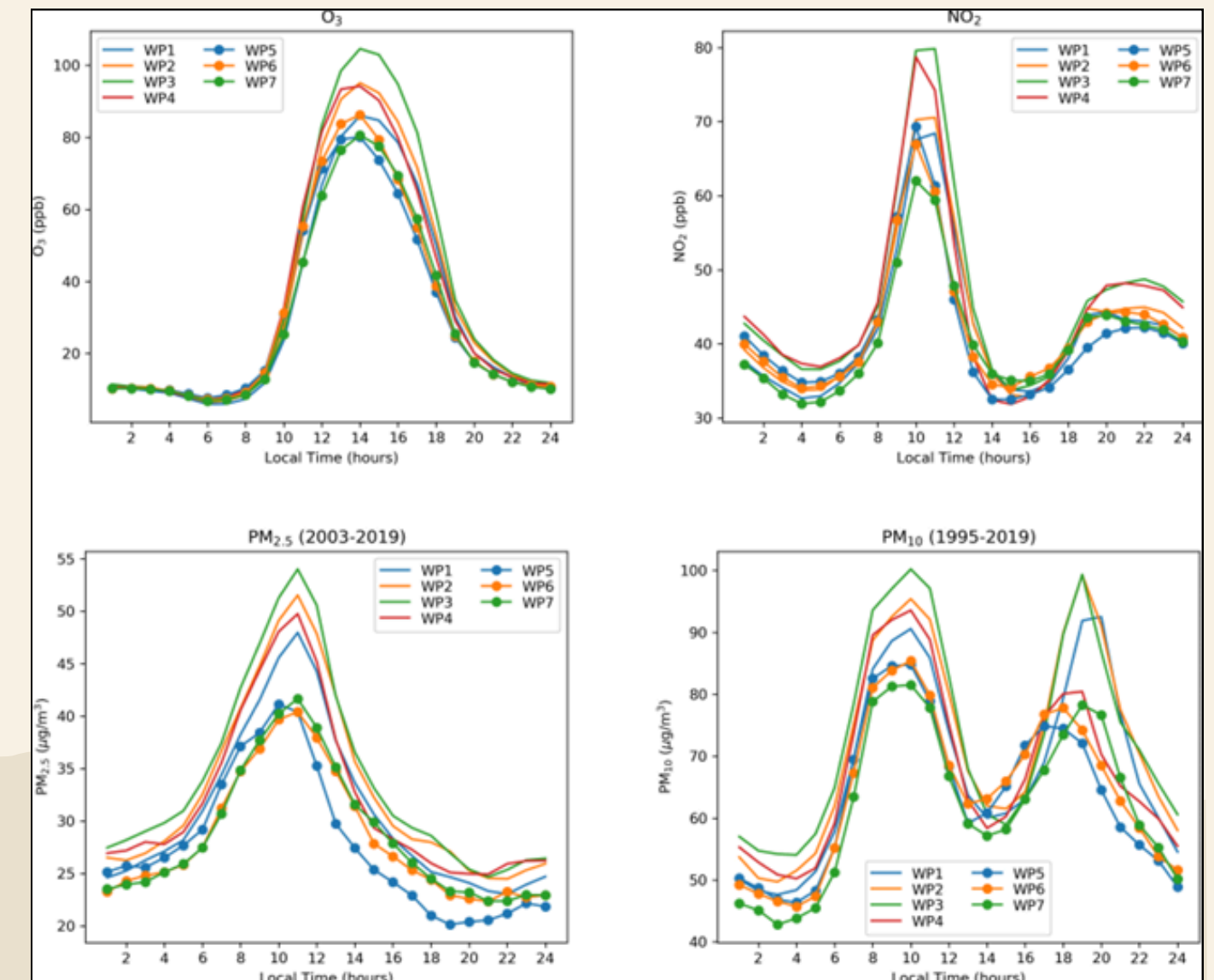
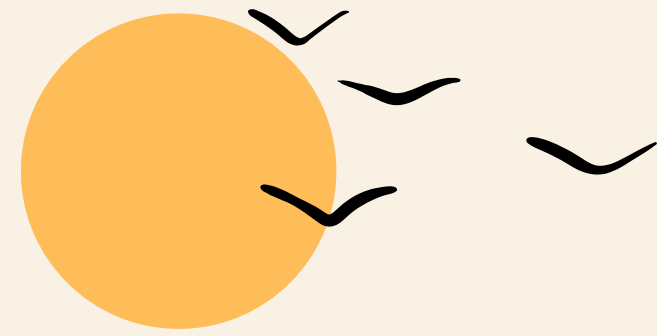


Fig. 10. Diurnal cycle of pollutants in the Metropolitan Area of Mexico City for each weather pattern. (Díaz-Esteban et al., 2019).

WEATHER PATTERNS & SEASONALITY (2/3)

Díaz-Esteban et al. (2022)



High PM10 and PM2.5 are caused by low wind speeds and atmospheric convergence (confluence line), which actively trap and accumulate pollutants.

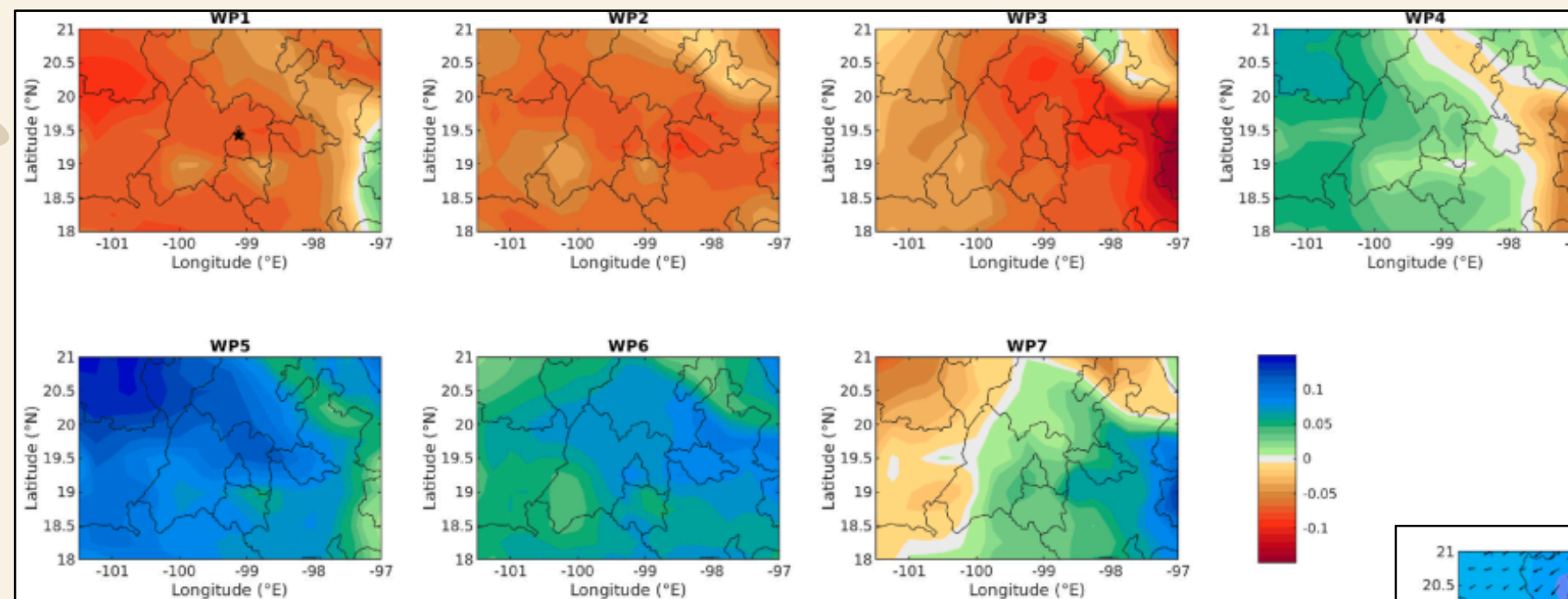


Fig. 11. Anomaly composites of total cloud cover over the MCMA, for each weather pattern. Black marker in WP1 indicates the location of Merced station (Díaz-Esteban et al., 2019).

Low PM levels are associated with mid-latitude troughing, which increases wind speeds and vertical mixing (dispersion) across the lower atmosphere.

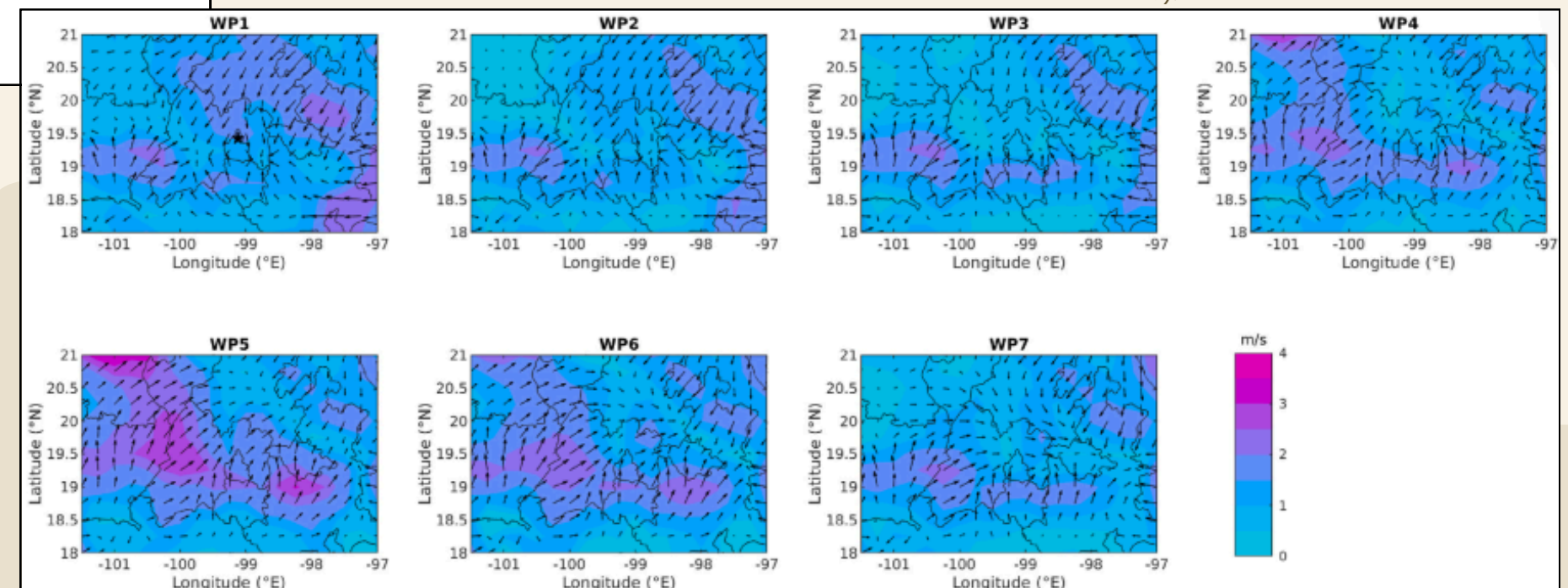


Fig. 12. Composites of mean 10-m wind (shading and vectors, in m/s) over the MCMA, for each weather pattern. Black marker in WP1 indicates the location of Merced station (Díaz-Esteban et al., 2019).

WEATHER PATTERNS & SEASONALITY (3/3)

Díaz-Esteban et al. (2022)

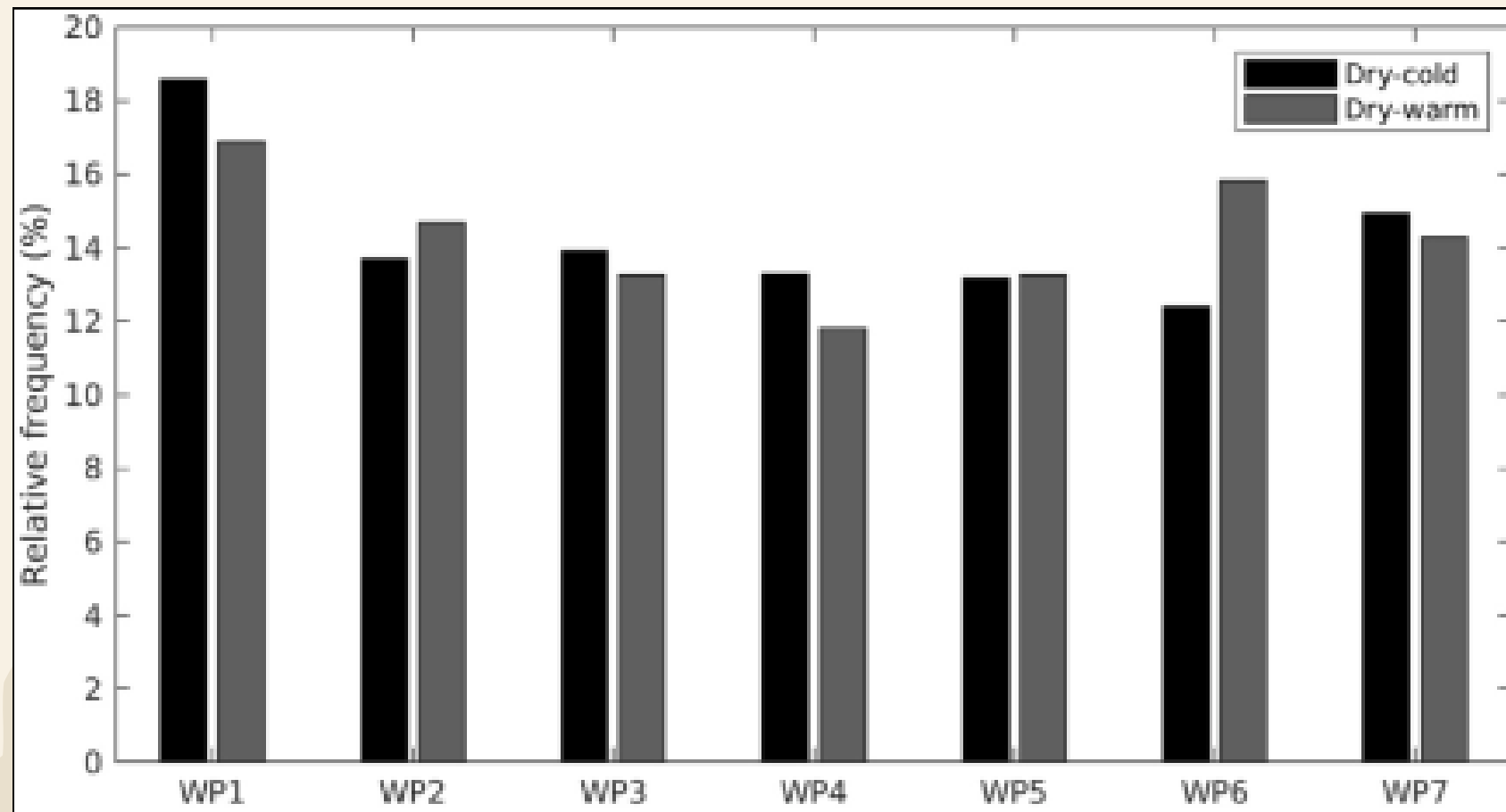
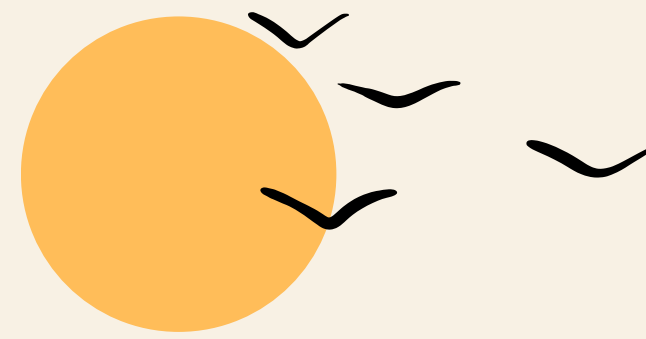


Fig. 13. Relative seasonal frequencies of each weather pattern. For each of the two seasons, the frequencies add up to 100% (Díaz-Esteban et al., 2019).

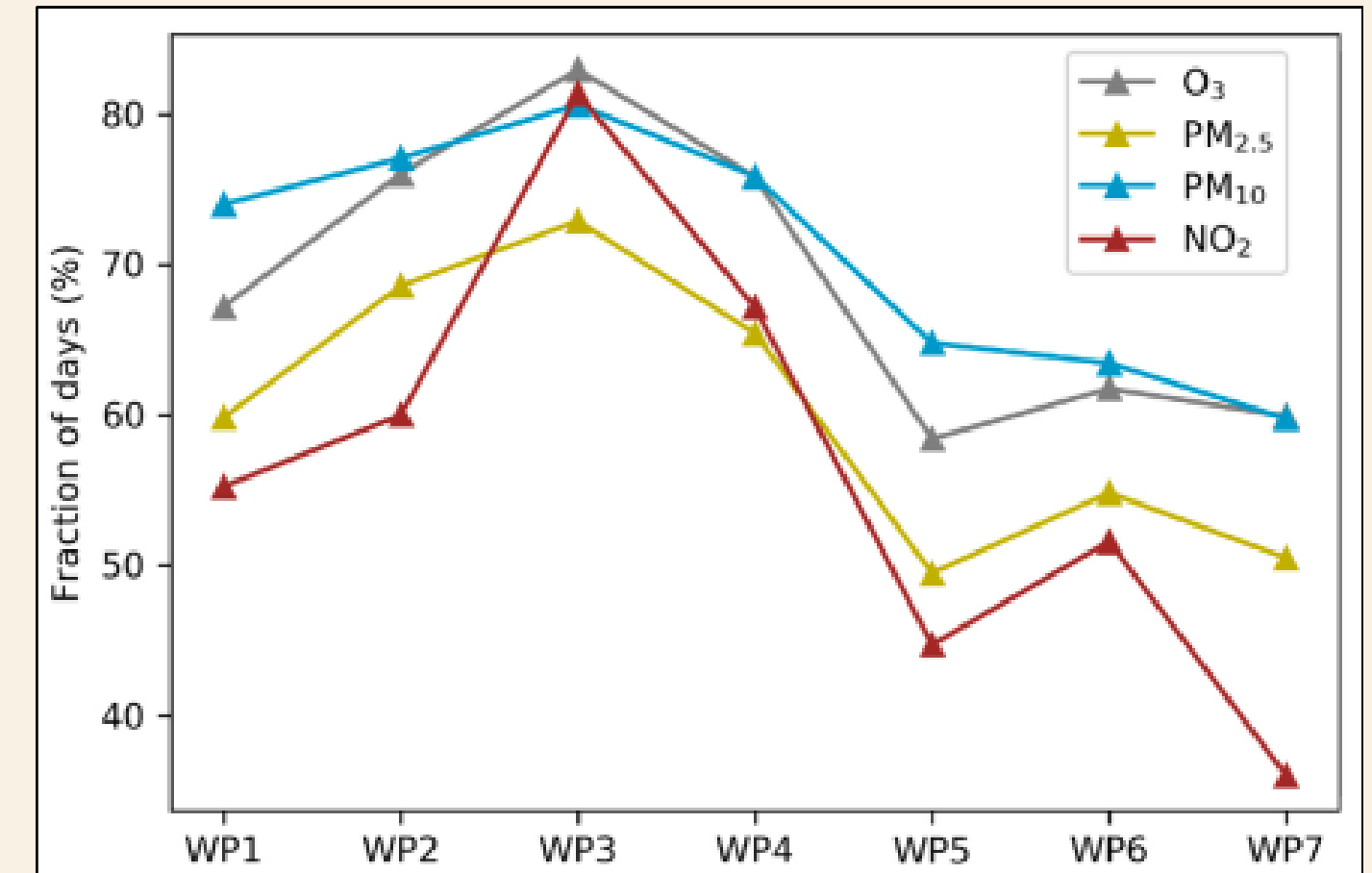
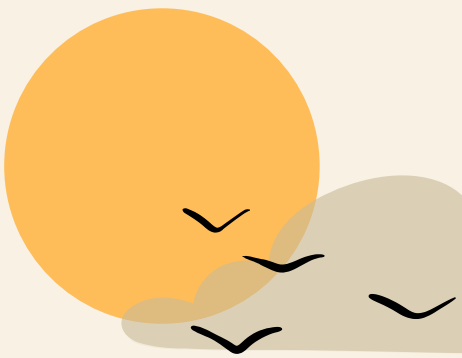


Fig. 14. Fraction of days where the concentration of pollutants exceeded the World Health Organization air quality guideline values, for each weather pattern (Díaz-Esteban et al., 2019).

- WP1 is the most common and persistent pattern. Occurrence varies seasonally: WP1, WP3, WP4 dominate the dry-cold season, while WP2, WP5, WP6 favor the dry-warm season.
- The Ridge patterns (WP2-WP4) cause the highest fraction of High Pollution Events (WP3 is the most polluted). Conversely, the Trough patterns (WP5-WP7) are consistently associated with the cleanest air.

CROSS-THEME INTEGRATION



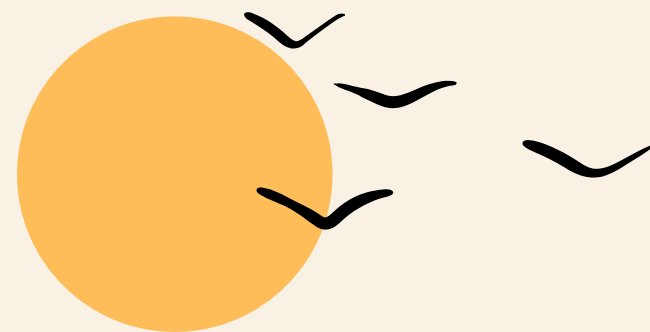
Dust raises PM, weather traps it, health studies quantify the damage, and policy translates it into public guidance.

Table 6. Cross-Theme Integration.

Theme	Mechanism	Effect	Policy / Health Implication
Dust Transport (June–July)	Saharan dust raises PM2.5/ PM10 levels 2–5x	Acute health risk; adds background PM	Highlight seasonal alerts in forecast systems
O ₃ Averaging 1 h vs 8 h	Focus on acute exposure communication	Clarifies risk messaging and aligns with WHO	Supports NOM-172 revision logic
Health Exposure Studies	Short-term pollution ↑ CEDVs (+10 %), long-term ↑ COVID mortality	Strengthens public health link	Quantify avoidable cases if pollution reduced 10 %
Weather Patterns	Synoptic blocking & mid-tropospheric high = stagnation → O ₃ peaks	Predictable seasonal events	Basis for forecasting and adaptation plans

- Dust → PM baseline increase in summer.
- O₃ policy (1-h) → clearer communication of acute risk.
- Short + long term exposure → hospital visits and mortality links.
- Weather patterns → predict when pollution episodes occur.
- Integration forms basis for forecast-based health alerts.





DISCUSSION & NEXT STEPS

- **Discussion:**

- This literature review clarified how dust transport, short-term exposure, and synoptic weather patterns jointly shape Mexico City's pollution episodes.
- The reviewed studies show that meteorology—not only emissions—controls the timing and intensity of pollution peaks.
- The NOM-172 shift to a 1-hour O_3 metric aligns with these findings, emphasizing short-term health communication.
- Integrating these insights helps interpret why certain months (March–May, June–July) consistently show poor air quality.

- **Next Steps (week 5 focus + realistic directions):**

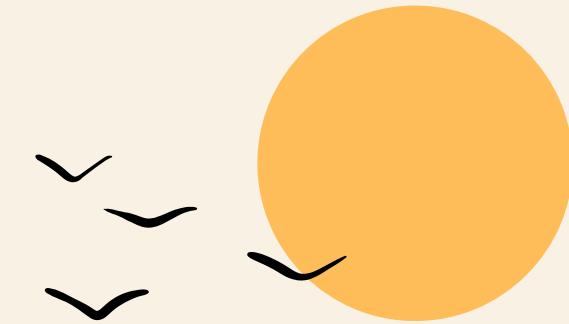
- Refine the temporal analysis:
 - Compare seasonal pollutant peaks in CDMX data (2012–2024) with the periods identified in the literature (MAM for O_3 , DJF for $PM_{2.5}$).
- Map meteorological context:
 - Overlay pollutant peaks with basic weather variables (temperature, RH, wind speed) from available stations or reanalysis.
- Identify potential indicators:
 - Note simple metrics (e.g., low wind + high temperature = likely O_3 spike) to explore in later analysis, without building a forecast model yet.

Understanding when and why pollution peaks occur is the foundation for any future predictive or policy work—simple, evidence-based patterns are already powerful insights.

Weather → Pollutant → Health → Policy



REFERENCES



1. Raga, G. B., et al., (2021): ADABBOY: African Dust And Biomass Burning Over Yucatan. *Bull. Amer. Meteor. Soc.*, 102, E1543–E1556. <https://doi.org/10.1175/BAMS-D-20-0172.1>
2. MapChart. (2025). Mexico | MapChart. <https://www.mapchart.net/mexico.html>
3. Kutralam-Muniasamy, G., Pérez-Guevara, F., Martínez, I.E. et al. (2021). Particulate matter concentrations and their association with COVID-19-related mortality in Mexico during June 2020 Saharan dust event. *Environ Sci Pollut Res* 28, 49989–50000. <https://doi.org/10.1007/s11356-021-14168-y>
4. Secretaría de Medio Ambiente y Recursos Naturales-SEMARNAT, (2023). Norma Oficial Mexicana NOM-172-SEMARNAT-2023, Lineamientos para la obtención y comunicación del índice de calidad del aire y riesgos a la salud. Retrieved from: <https://rama.edomex.gob.mx/sites/rama.edomex.gob.mx/files/files/NOM-172-SEMARNAT-2023.pdf>
5. Ugalde-Resano R., Riojas-Rodríguez H., Texcalac-Sangrador J. L., Cruz J.C., & Hurtado-Díaz M. (2022). Short term exposure to ambient air pollutants and cardiovascular emergency department visits in Mexico city. *Environmental Research*, Volume 207, 112600. <https://doi.org/10.1016/j.envres.2021.112600>
6. López-Feldman A., Heres D. & Márquez-Padilla F. (2021). Air pollution exposure and COVID-19: A look at mortality in Mexico City using individual-level data. *Science of The Total Environment*, Volume 756, 143929. <https://doi.org/10.1016/j.scitotenv.2020.143929>
7. Cromar, K., Gladson, L., Jaimes Palomera, M., & Perlmutter, L. (2021). Development of a Health-Based Index to Identify the Association between Air Pollution and Health Effects in Mexico City. *Atmosphere*, 12(3), 372. <https://doi.org/10.3390/atmos12030372>
8. Salcido, A., Carreón-Sierra, S., & Celada-Murillo, A. T. (2019). Air Pollution Flow Patterns in the Mexico City Region. *Climate*, 7(11), 128. <https://doi.org/10.3390/cli7110128>
9. Silva-Quiroz R., Rivera A. L., Ordoñez P., Gay-García C., & Frank A. (2019). Atmospheric blockages as trigger of environmental contingencies in Mexico City. *Heliyon*, 5(e02099). <https://doi.org/10.1016/j.scitotenv.2020.143929>
10. Díaz-Esteban Y., Barret B. S., Ordoñez P., & Raga G. B. (2022). Circulation patterns influencing the concentration of pollutants in central Mexico. *Atmospheric Environment*, 274(118976). <https://doi.org/10.1016/j.scitotenv.2020.143929>



THANK YOU!



SEE4994
RICO ESPARZA, Erick
17/10/2025

